



Lab#13

FULL STACK WEB DEVELOPMENT

BSSE- 6B

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Presented to:

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Github Link for All the tasks: <https://github.com/RajaaTayyab/Full-Stack-Programming-Lab>

API RESTful Deployment and Testing -Lab

Task 1: Lab Weather Forecast API (Real City-Based System)

Design and develop a REST API using Node.js and Express.js that integrates with an external weather service to provide real-time weather information for different cities.

Task Requirements:

- Create a REST API endpoint that accepts a city name as a parameter
- Connect your API with a third-party weather service (e.g., OpenWeather API)
- <https://openweathermap.org/api>
- Return weather information in JSON format

The response must include:

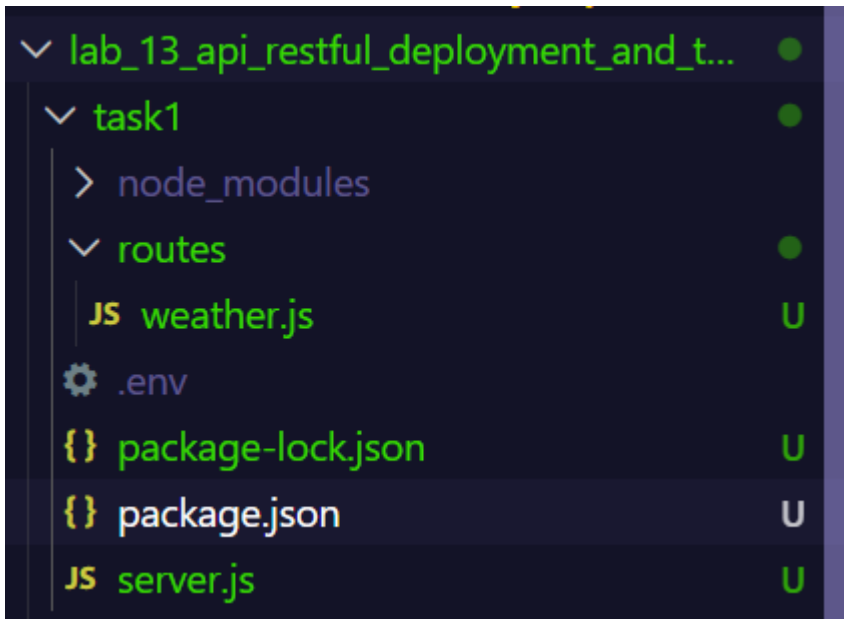
- City name
- Current temperature
- Weather condition (e.g., cloudy, sunny, rain)
- Humidity level
- Handle errors properly (e.g., invalid city name or API failure)

Test your API using Postman or browser

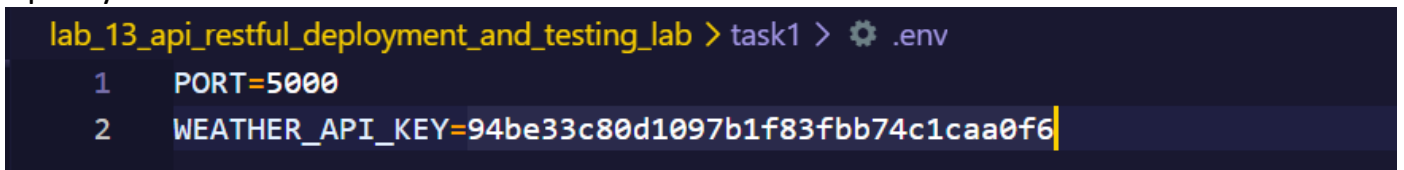
- It should be able to fetch live weather data for any

Screenshots

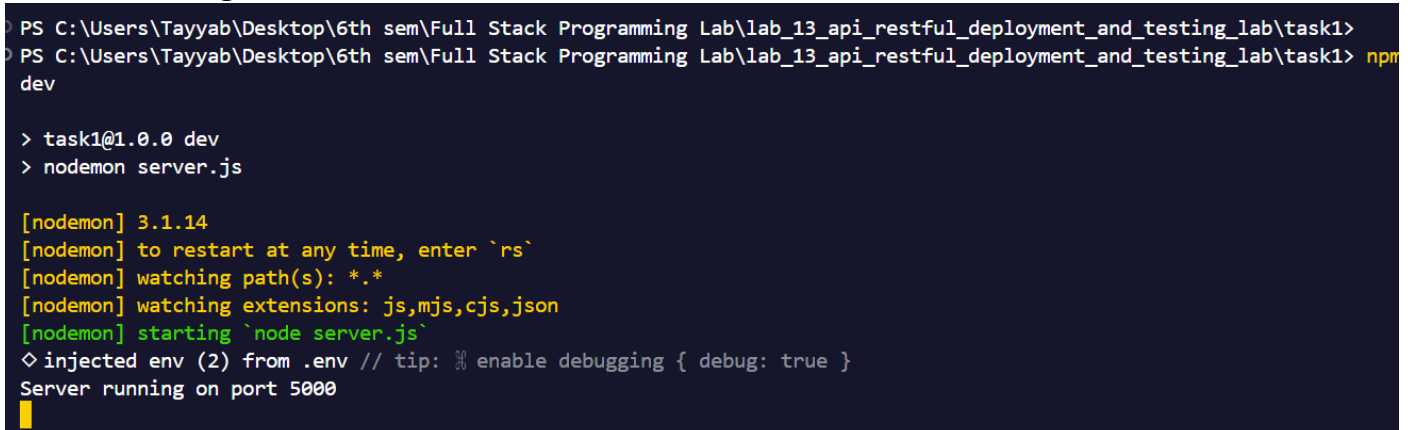
Project Structure



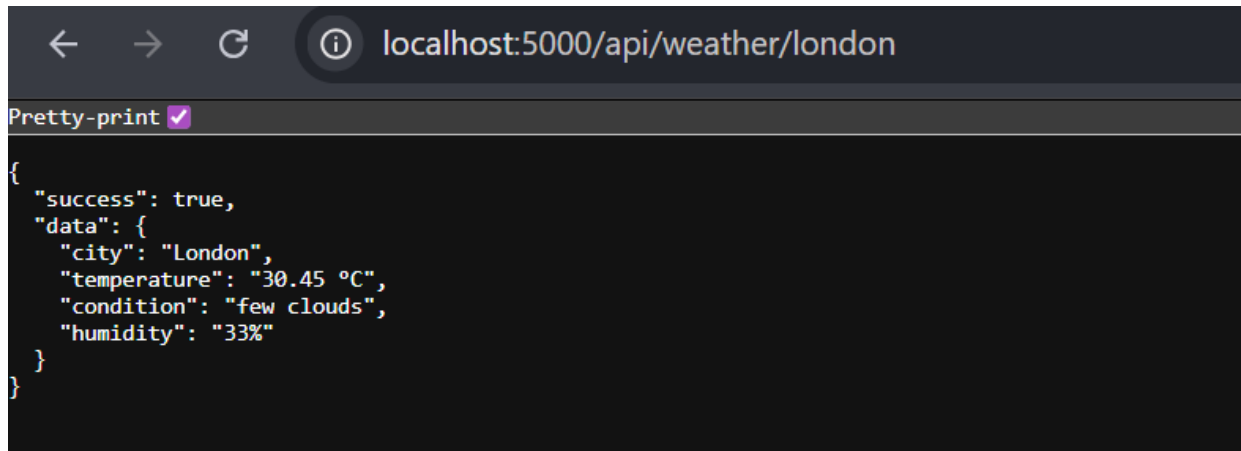
Api key



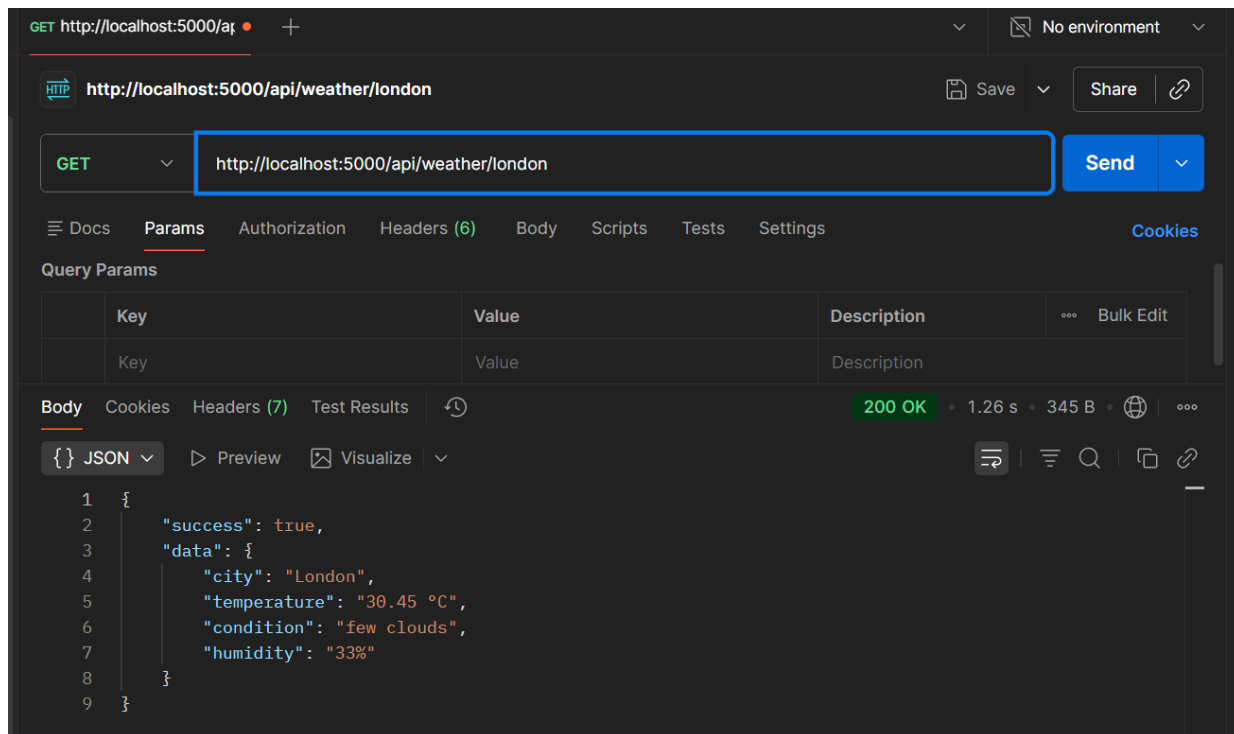
Server Running



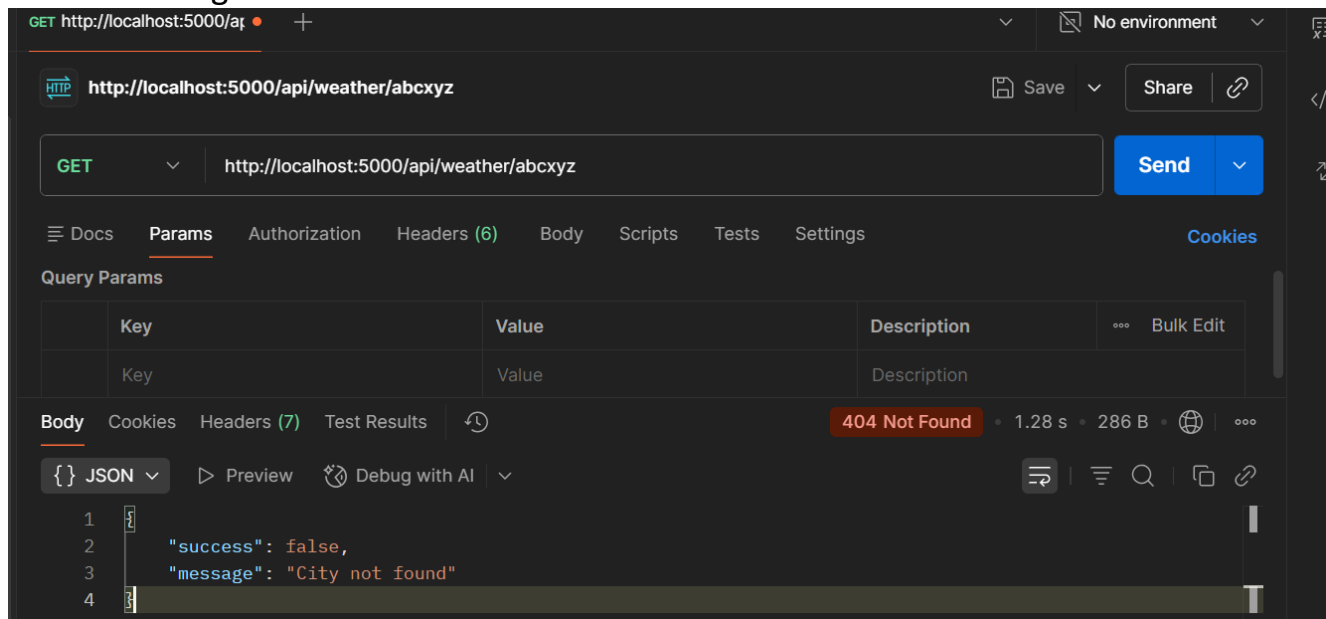
Browser Test



Postman Test



Error Handling



Task 2: News Headlines API using External Service

To develop a RESTful API using Node.js and Express.js that fetches and displays the latest news headlines using a third-party news service.

- Work with external REST APIs (<https://newsapi.org/>)
- Handle dynamic route parameters
- Process and filter API responses
- Build structured JSON responses
- Implement error handling in backend applications

Design and implement a REST API that retrieves the latest news headlines for a given country using

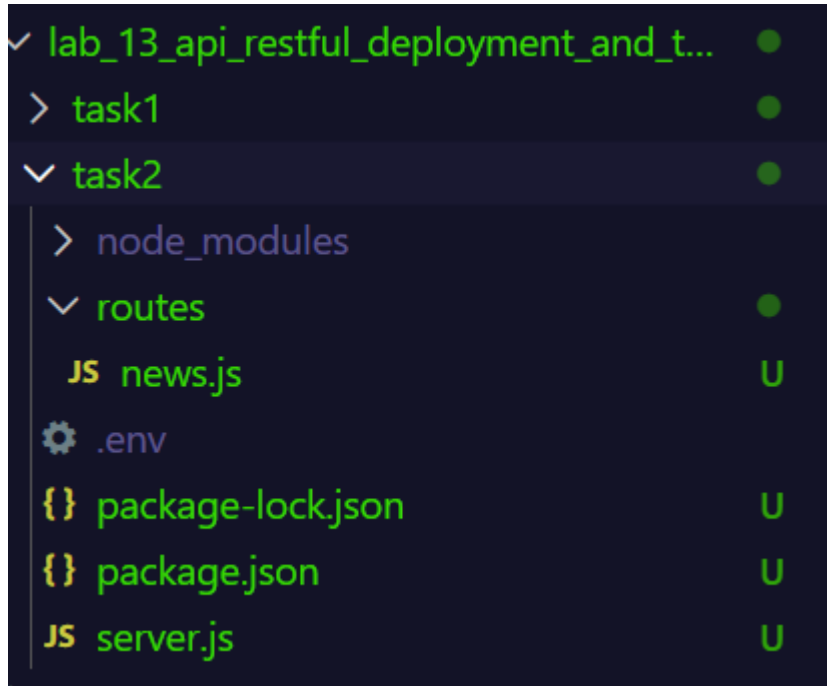
an external news API service.

Requirements

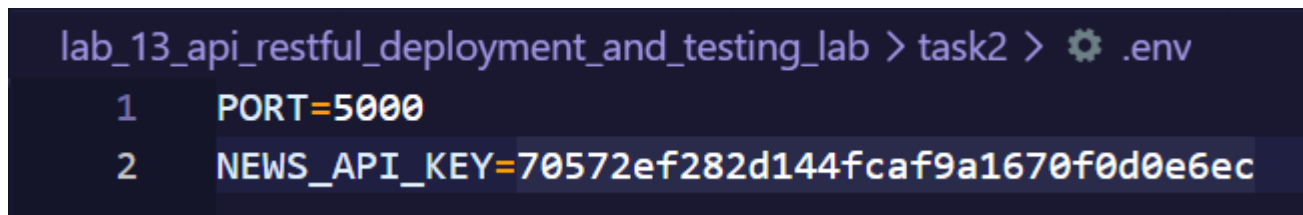
- Accept country code as a URL parameter
- Connect with a third-party news API service
- Fetch latest top headlines
- Return structured JSON format
- Include: News Title, Source Name, News URL, Publication Date
- Limit response to 5–10 articles
- Handle API errors and invalid inputs
- Test using Postman and browser
- A working REST API that provides real-time top headlines for a selected country.

Screenshots

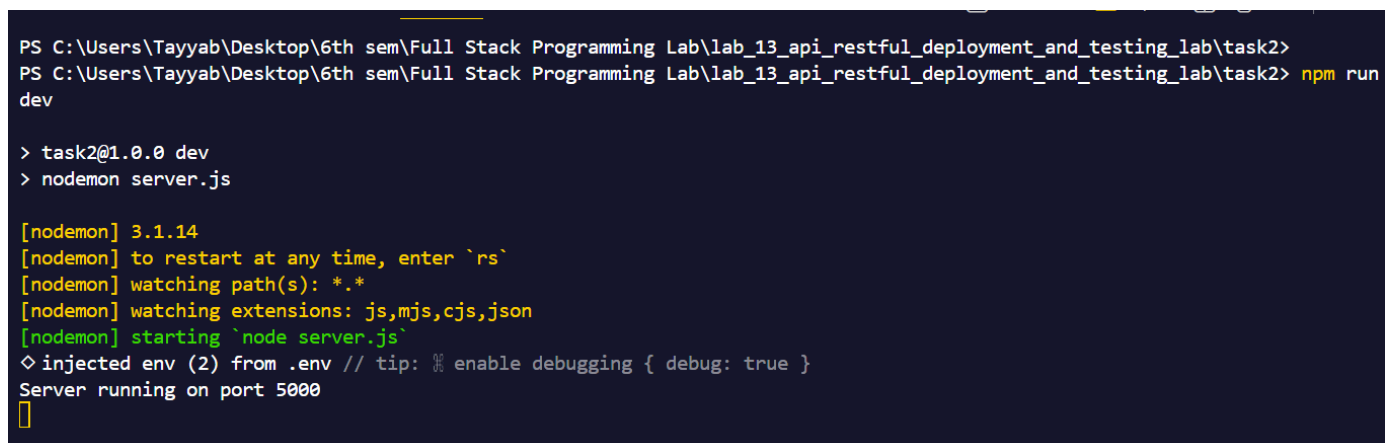
Project Structure



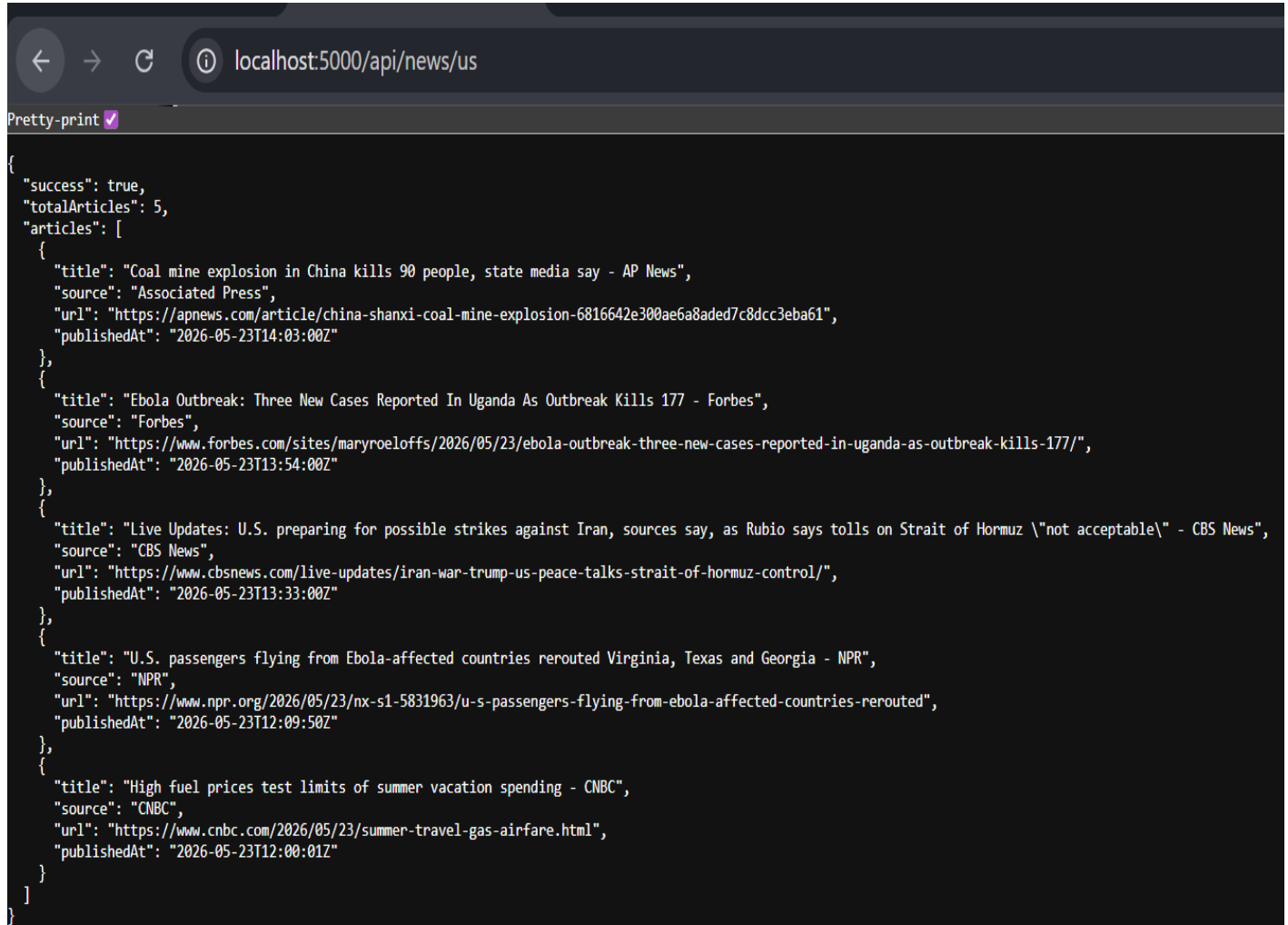
Api key



Server Running



Browser Test



```
{
  "success": true,
  "totalArticles": 5,
  "articles": [
    {
      "title": "Coal mine explosion in China kills 90 people, state media say - AP News",
      "source": "Associated Press",
      "url": "https://apnews.com/article/china-shanxi-coal-mine-explosion-6816642e300ae6a8aded7c8dcc3eba61",
      "publishedAt": "2026-05-23T14:03:00Z"
    },
    {
      "title": "Ebola Outbreak: Three New Cases Reported In Uganda As Outbreak Kills 177 - Forbes",
      "source": "Forbes",
      "url": "https://www.forbes.com/sites/maryroeloffs/2026/05/23/ebola-outbreak-three-new-cases-reported-in-uganda-as-outbreak-kills-177/",
      "publishedAt": "2026-05-23T13:54:00Z"
    },
    {
      "title": "Live Updates: U.S. preparing for possible strikes against Iran, sources say, as Rubio says tolls on Strait of Hormuz \"not acceptable\" - CBS News",
      "source": "CBS News",
      "url": "https://www.cbsnews.com/live-updates/iran-war-trump-us-peace-talks-strait-of-hormuz-control/",
      "publishedAt": "2026-05-23T13:33:00Z"
    },
    {
      "title": "U.S. passengers flying from Ebola-affected countries rerouted Virginia, Texas and Georgia - NPR",
      "source": "NPR",
      "url": "https://www.npr.org/2026/05/23/nx-s1-5831963/u-s-passengers-flying-from-ebola-affected-countries-rerouted",
      "publishedAt": "2026-05-23T12:09:50Z"
    },
    {
      "title": "High fuel prices test limits of summer vacation spending - CNBC",
      "source": "CNBC",
      "url": "https://www.cnbc.com/2026/05/23/summer-travel-gas-airfare.html",
      "publishedAt": "2026-05-23T12:00:01Z"
    }
  ]
}
```

Postman Test

GET http://localhost:5000/api/news/us

HTTP http://localhost:5000/api/news/us Save Share

GET http://localhost:5000/api/news/us Send

Docs Params Authorization Headers (6) Body Scripts Tests Settings Cookies

Query Params

Key	Value	Description
Key	Value	Description

Body Cookies Headers (7) Test Results 200 OK • 683 ms • 1.57 KB

{ } JSON Preview Visualize

```
1 {
2   "success": true,
3   "totalArticles": 5,
4   "articles": [
5     {
6       "title": "Coal mine explosion in China kills 90 people, state media say - AP News",
7       "source": "Associated Press",
8       "url": "https://apnews.com/article/
9         china-shanxi-coal-mine-explosion-6816642e300ae6a8aded7c8dcc3eba61",
10      "publishedAt": "2026-05-23T14:03:00Z"
11    },
12    {
13      "title": "Ebola Outbreak: Three New Cases Reported In Uganda As Outbreak Kills 177 - Forbes",
14      "source": "Forbes",
15      "url": "https://www.forbes.com/sites/maryroeloffs/2026/05/23/
16        ebola-outbreak-three-new-cases-reported-in-uganda-as-outbreak-kills-177/",
17      "publishedAt": "2026-05-23T13:54:00Z"
18    },
19    {
20      "title": "Live Updates: U.S. preparing for possible strikes against Iran, sources say, as Rubio says tolls on Strait of Hormuz \"not acceptable\" - CBS News",
21      "source": "CBS News",
22      "url": "https://www.cbsnews.com/live-updates/
23        iran-war-trump-us-peace-talks-strait-of-hormuz-control/",
24      "publishedAt": "2026-05-23T13:33:00Z"
25    },
26    {
27      "title": "High fuel prices test limits of summer vacation spending - CNBC",
28      "source": "CNBC",
29      "url": "https://www.cnbc.com/2026/05/23/summer-travel-gas-airfare.html",
30      "publishedAt": "2026-05-23T12:00:01Z"
31    }
32  ]
33 }
```

```
26    {
27      "title": "Ebola Outbreak: Three New Cases Reported In Uganda As Outbreak Kills 177 - Forbes",
28      "source": "Forbes",
29      "url": "https://www.forbes.com/sites/maryroeloffs/2026/05/23/
30        ebola-outbreak-three-new-cases-reported-in-uganda-as-outbreak-kills-177/",
31      "publishedAt": "2026-05-23T13:54:00Z"
32    },
33    {
34      "title": "Live Updates: U.S. preparing for possible strikes against Iran, sources say, as Rubio says tolls on Strait of Hormuz \"not acceptable\" - CBS News",
35      "source": "CBS News",
36      "url": "https://www.cbsnews.com/live-updates/
37        iran-war-trump-us-peace-talks-strait-of-hormuz-control/",
38      "publishedAt": "2026-05-23T13:33:00Z"
39    },
40    {
41      "title": "High fuel prices test limits of summer vacation spending - CNBC",
42      "source": "CNBC",
43      "url": "https://www.cnbc.com/2026/05/23/summer-travel-gas-airfare.html",
44      "publishedAt": "2026-05-23T12:00:01Z"
45    }
46  ]
47 }
```

```
28   },
29   {
30     "title": "High fuel prices test limits of summer vacation spending - CNBC",
31     "source": "CNBC",
32     "url": "https://www.cnbc.com/2026/05/23/summer-travel-gas-airfare.html",
33     "publishedAt": "2026-05-23T12:00:01Z"
34   }
35 ]
36 }
```


Error Handling

The screenshot shows a REST client interface with a GET request to `http://localhost:5000/api/news/xyz`. The response is a **400 Bad Request** with a status of 85 ms and 294 B. The response body is JSON, showing an error message: `"message": "Invalid country code"`.

```
{  "success": false,  "message": "Invalid country code"}
```

Error Handling on Browser

The screenshot shows a web browser address bar with the URL `localhost:5000/api/news/xyz`. Below the address bar, the response is displayed in a pretty-printed JSON format, showing the same error message: `"message": "Invalid country code"`.

```
{  "success": false,  "message": "Invalid country code"}
```